

bldc_controller_routed

Design Report

Rev 1 | 2026-09-10 | jlcpcb

kicad-tools 0.20.0

Board Summary

Property	Value
Layers	4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu)
Footprints	42 (36 SMD, 6 THT, 0 other)
Nets	37
Traces	7983 segments
Vias	136
Board Size	70.0 x 90.0 mm

Stackup

Layer	Type	Thickness (mm)	Material
F.Mask	Top Solder Mask	0.01	--
F.Cu	copper	0.035	--
dielectric 1	prepreg	0.2	FR4
In1.Cu	copper	0.035	--
dielectric 2	core	1.0	FR4
In2.Cu	copper	0.035	--
dielectric 3	prepreg	0.2	FR4
B.Cu	copper	0.035	--
B.Mask	Bottom Solder Mask	0.01	--

Design Overview

Theory of Operation

Sensored BLDC Controller

Communication Interfaces

Protocol	Signals
SPI	MISO, MOSI, SCK

Power Architecture

Power Rails: PWR_FLAG

Assembly Notes

1 fine-pitch component; 2 thermal pads; 4 polarized components

- **Fine-pitch components:** 1 (U1)

- **Thermal pads:** 2 -- verify solder paste apertures
- **Polarized components:** 4 -- check orientation markings

Hand-Solder (THT) Components

The following 6 through-hole components are **excluded from the SMT pick-and-place file** and must be hand-soldered (or wave/selective-soldered) after SMT assembly. They appear in the BOM for sourcing.

Value	Package	Qty	References
12-24V INPUT	PinHeader_1x02_P2.54 mm_Vertical	1	J1
MOTOR ABC	PinHeader_1x03_P2.54 mm_Vertical	1	J2
HALL	PinHeader_1x06_P2.54 mm_Vertical	1	J3
AVR ISP	PinHeader_2x03_P2.54 mm_Vertical	1	J4
RUN SWITCH	PinHeader_1x02_P2.54 mm_Vertical	1	J5
10k	Potentiometer_Bourns_ 3296W_Vertical	1	RV1

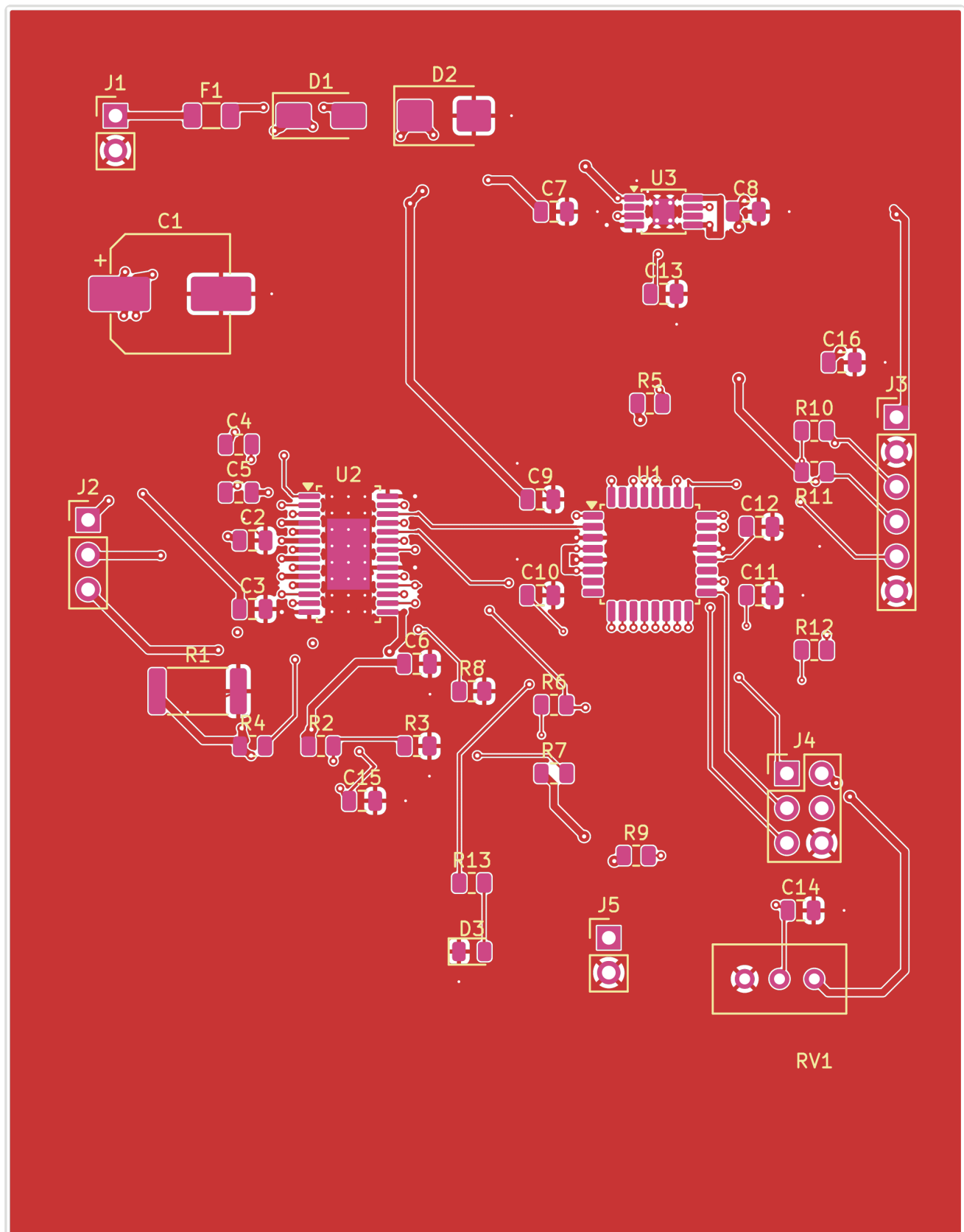
ERC Status

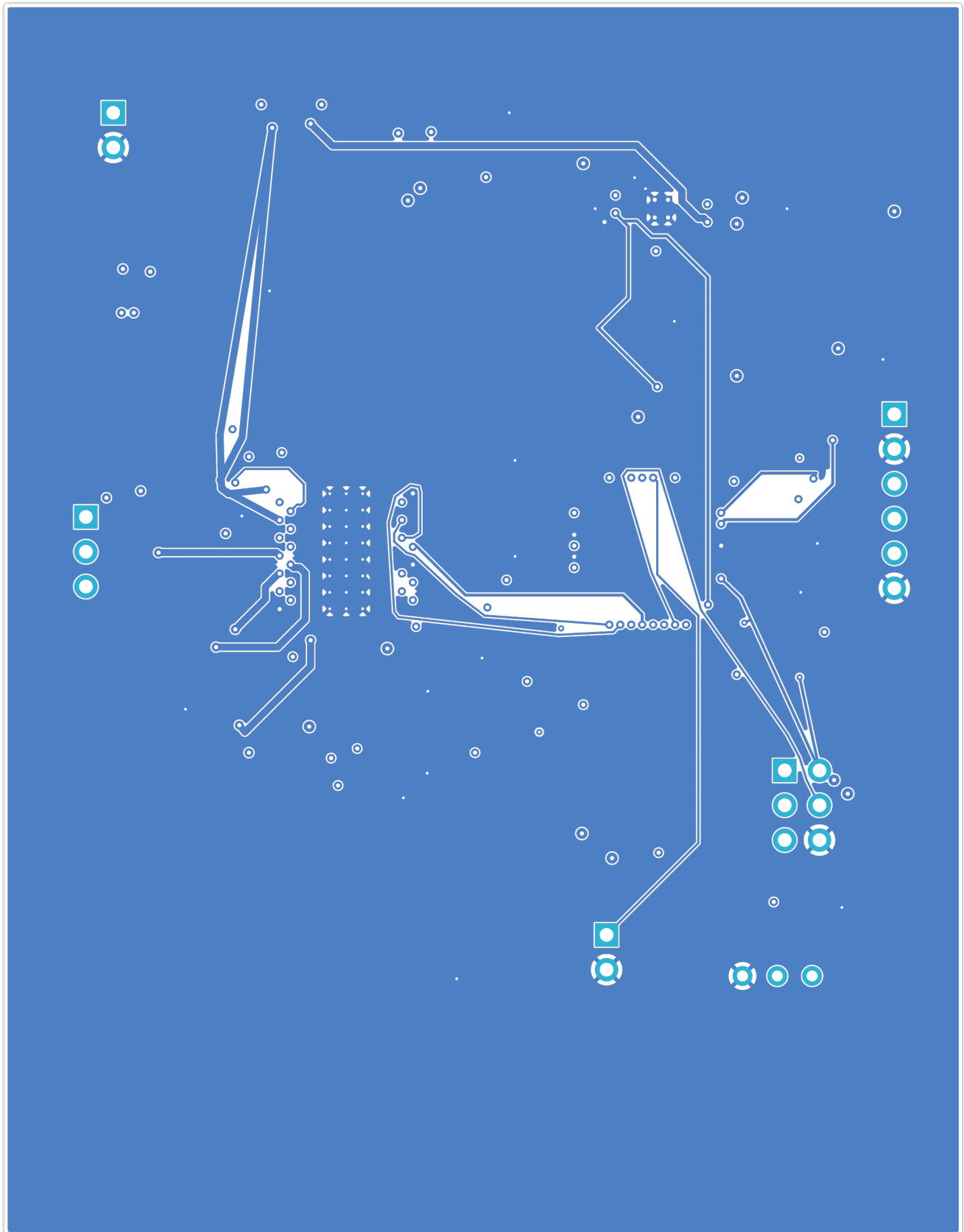
Metric	Count
Errors	0
Warnings	0

Status: SKIPPED -- ERC skipped by user request

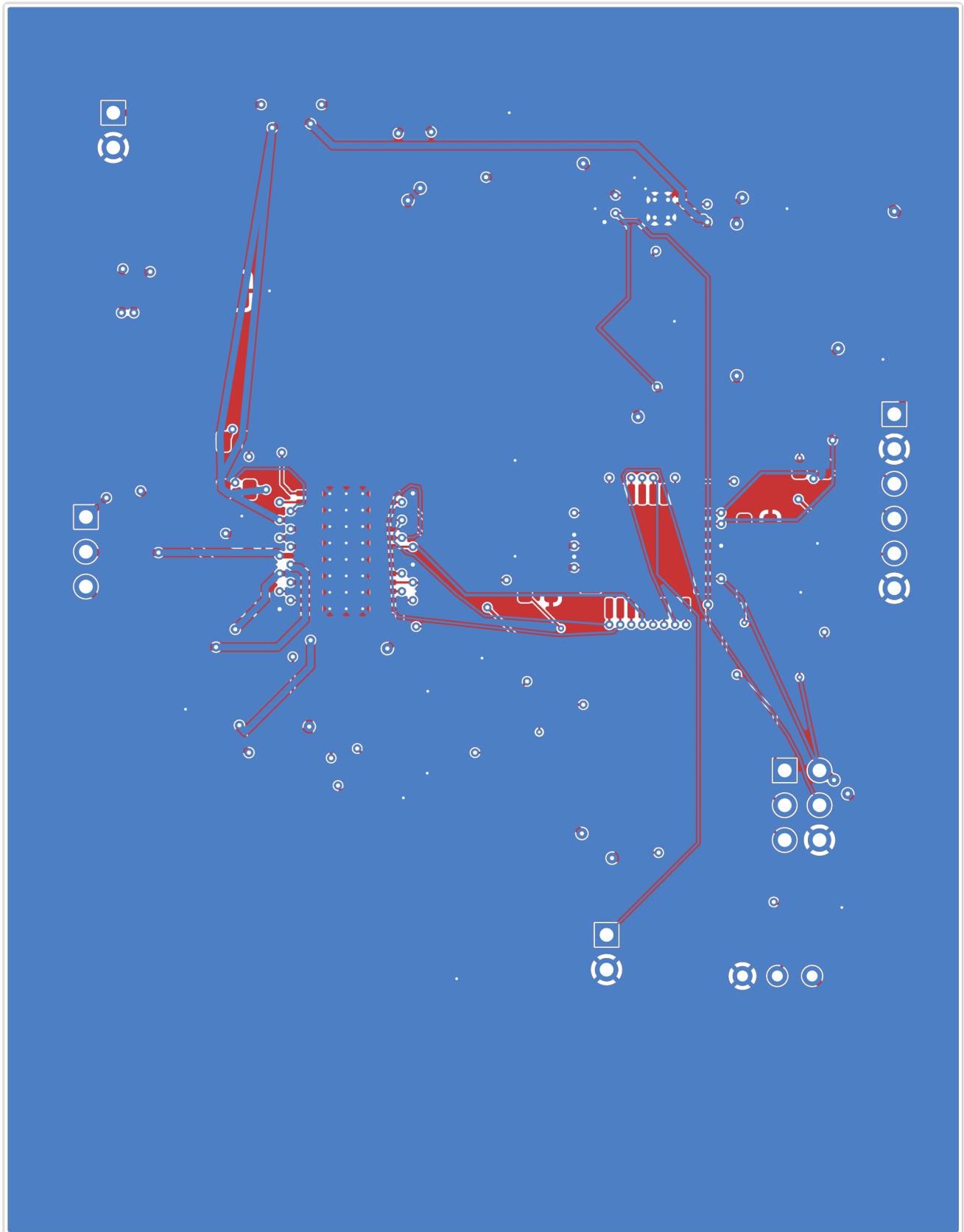
The PCB layout for the Sensored BLDC Controller is shown. The board is populated with various components including an ATmega328P-AU microcontroller, a 28C64 EEPROM, a 2N7000 MOSFET, a 555 timer, and numerous passive components like resistors and capacitors. The layout includes a header for a 12-24V input, a 5V regulator, and a 5-pin header for a BLDC motor. The PCB is labeled with component values and part numbers.

PCB Layout

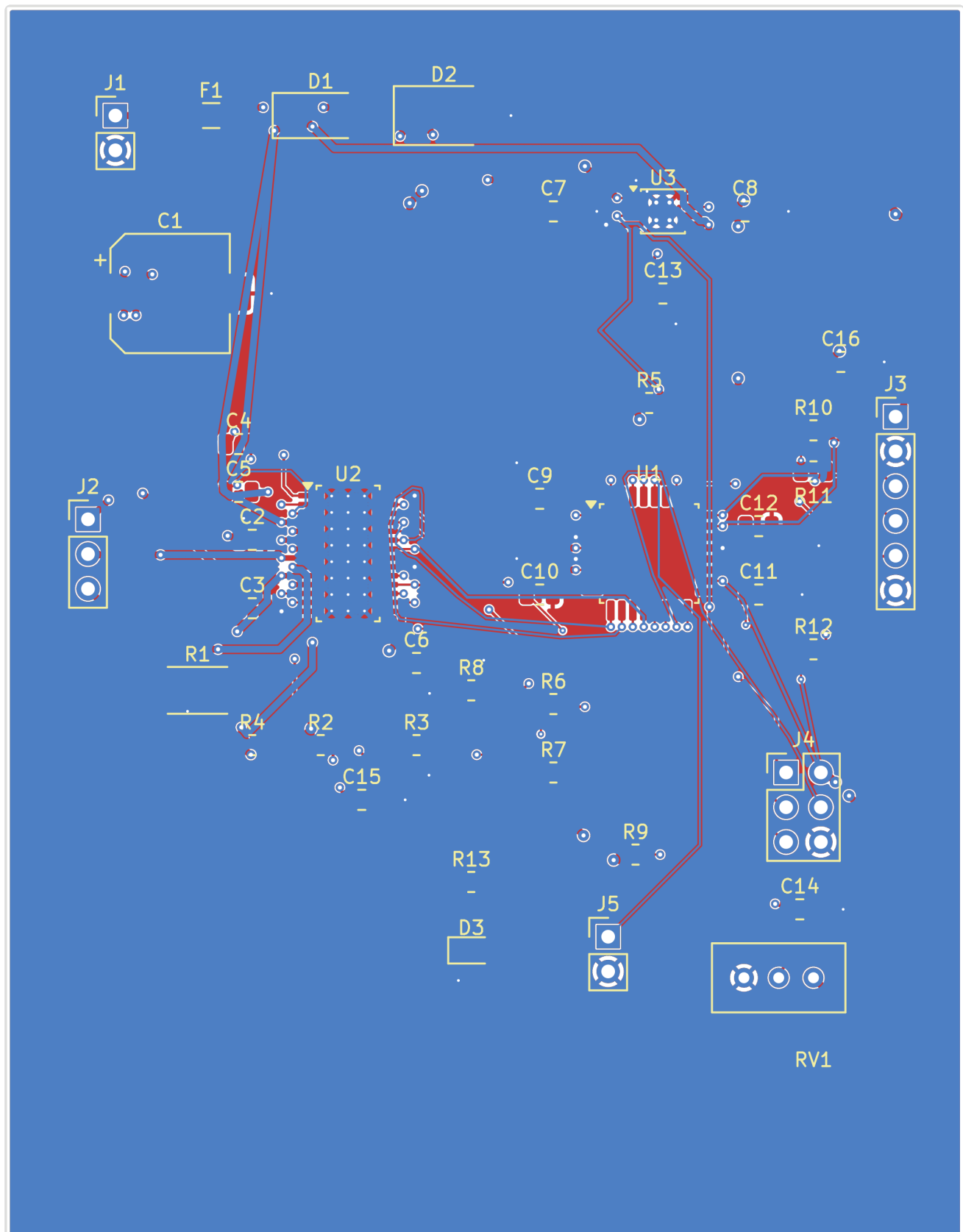




Copper

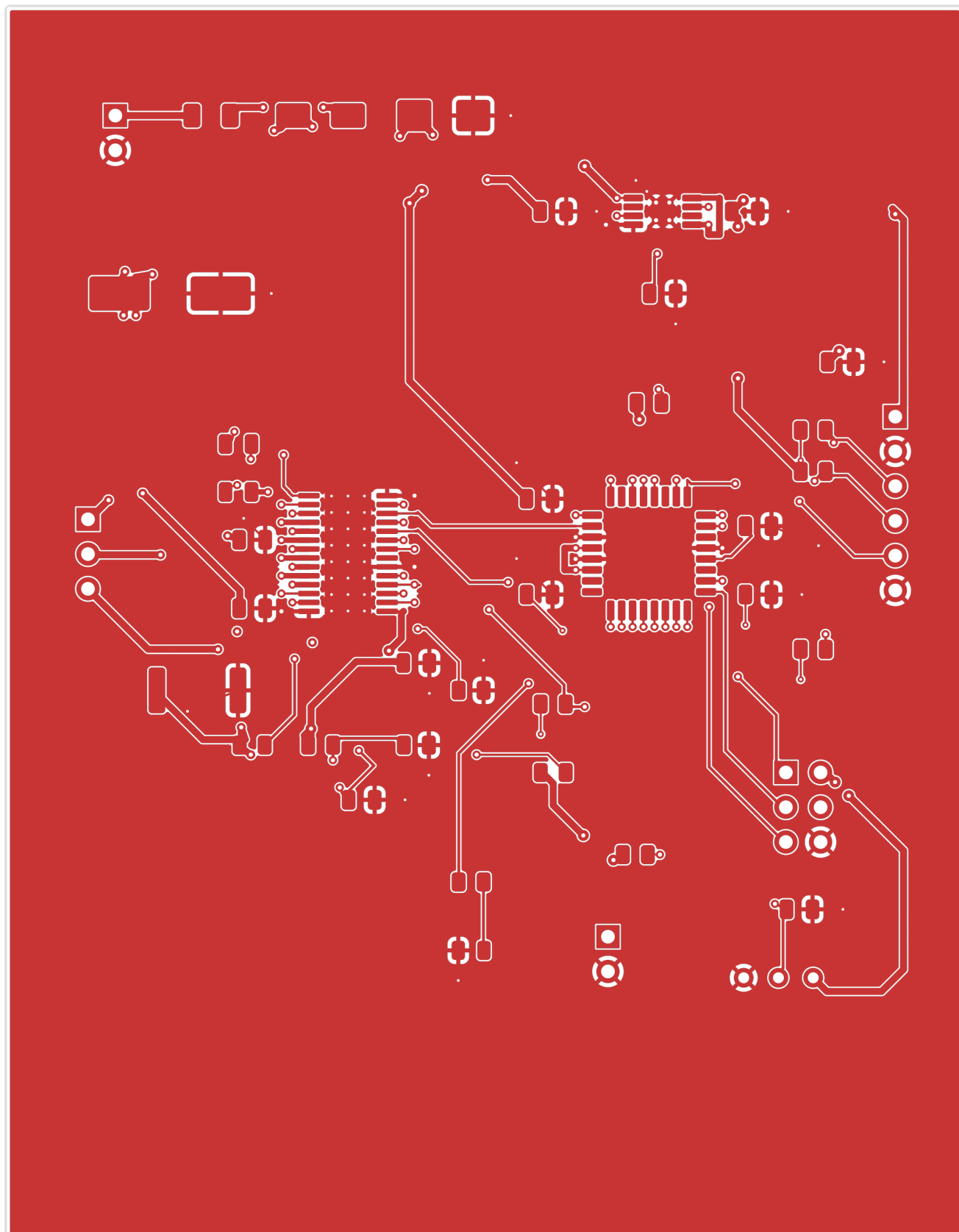


Assembly

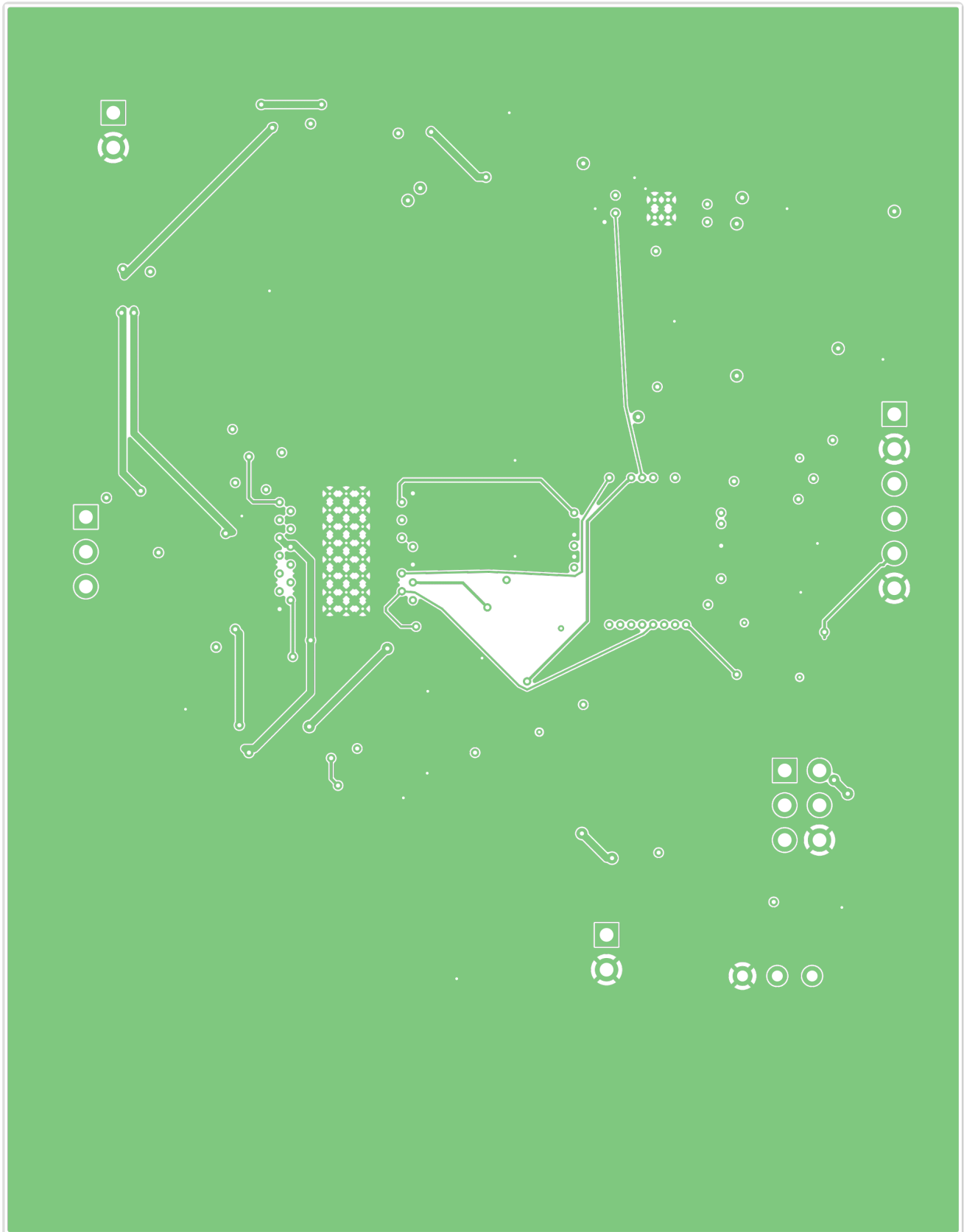


Copper Layers

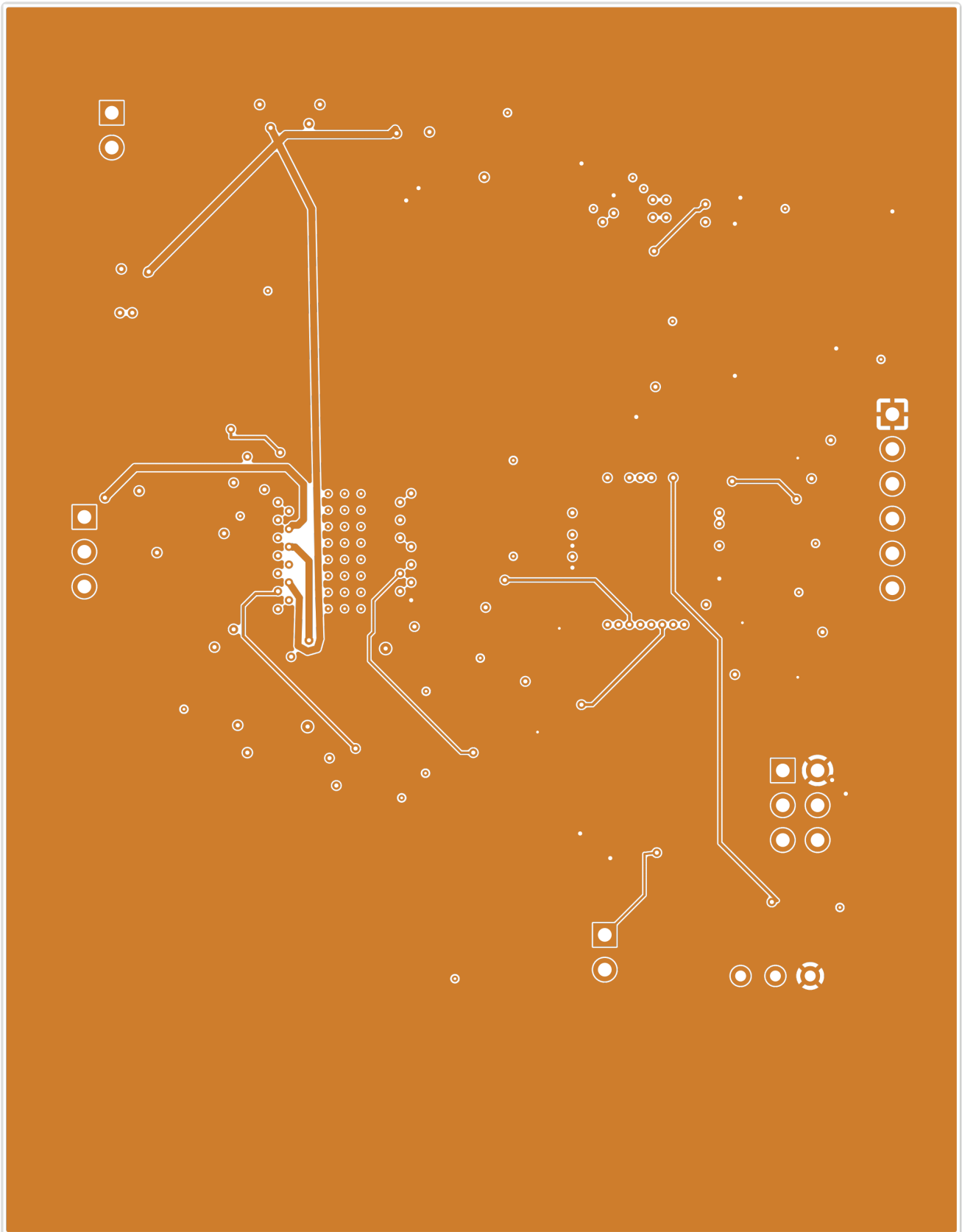
F.Cu



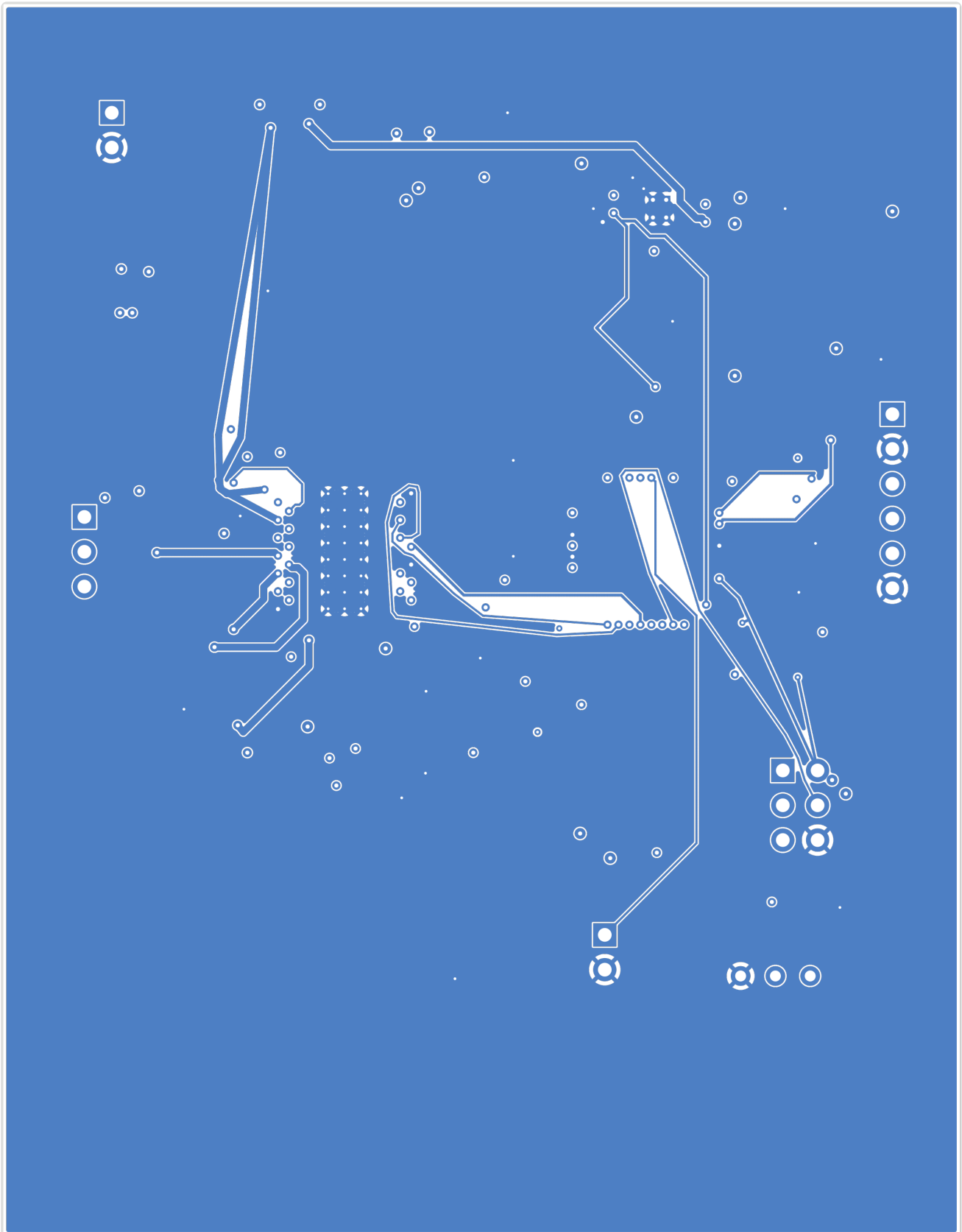
In1.Cu



In2.Cu



B.Cu



Bill of Materials

Value	Package	Qty	References	MPN	LCSC
100nF	C_0805_2012 Metric	10	C2, C3, C5, C9, C10, C11, C12, C14, C15, C16	CC0805KRX7R 9BB104	C49678
10nF	C_0805_2012 Metric	2	C4, C13	CL21B103KBA NNNC	C1710
10uF	C_0805_2012 Metric	1	C8	CL21A106KAY NNNE	C15850
1uF	C_0805_2012 Metric	1	C7	CL21B105KBF NNNE	C28323
220uF 50V	CP_Elec_8x10. 5_PolarityMark	1	C1	EEEFT1H221A P	C178594
470nF	C_0805_2012 Metric	1	C6	CL21B474KBF NNNE	C13967
RED	LED_0805_201 2Metric	1	D3	17-21SURC/ S530-A2/TR8	C131244
SMBJ24A	D_SMB	1	D2	SMBJ24A	C224017
SS36	D_SMA	1	D1	SS36	C16015
1A 63V	Fuse_1206_32 16Metric	1	F1	SF-1206S100- 2	C3167176
12-24V INPUT	PinHeader_1x0 2_P2.54mm_V ertical	1	J1	ZX- PZ2.54-1-2PZ Z	C7501260
AVR ISP	PinHeader_2x0 3_P2.54mm_V ertical	1	J4	2.54-2*3P	C65114
HALL	PinHeader_1x0 6_P2.54mm_V ertical	1	J3	2.54-1*6P	C37208
MOTOR ABC	PinHeader_1x0 3_P2.54mm_V ertical	1	J2	2.54-1*3P	C49257
RUN SWITCH	PinHeader_1x0 2_P2.54mm_V ertical	1	J5	ZX- PZ2.54-1-2PZ Z	C7501260

Value	Package	Qty	References	MPN	LCSC
0.5R	R_2512_6332 Metric	1	R1	WSL2512R500 0FEA	C511023
10k	Potentiometer_ Bourns_3296W _Vertical	1	RV1	3296W-1-103L F	C34846
10k	R_0805_2012 Metric	9	R3, R5, R6, R7, R8, R9, R10, R11, R12	0805W8F1002 T5E	C17414
1k	R_0805_2012 Metric	2	R4, R13	0805W8F1001 T5E	C17513
56k	R_0805_2012 Metric	1	R2	0805W8F5602 T5E	C17756
ATmega328P- AU	TQFP-32_7x7 mm_P0.8mm	1	U1	ATMEGA328P- AU	C14877
DRV8313PWP R	TI_PWP0028C _EP3.4x9.7_M ask3.1x5.18_T hermalVias	1	U2	DRV8313PWP R	C92482
TPS7A1650DG NR	HVSSOP-8-1E P_3x3mm_P0. 65mm_EP1.57 x1.89mm_Ther malVias	1	U3	TPS7A1650DG NR	C468238

DRC Status

Metric	Count
Errors	0
Warnings	0
Blocking	0

Status: PASS

Manufacturing Readiness

Verdict: WARNING

Action Items

- **[OPTIONAL]** Verify zone fill in KiCad for 2 zone-connected nets
- **[OPTIONAL]** Analog net: GND_SENSE — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: SENSE_IN — analog signal; noise-sensitive, avoid crossing digital signals

Routing Status

Metric	Value
Signal Net Completion	100.0% (35/35)
Overall Completion	100.0%
Complete Nets	37 / 37
Zone-Connected Nets	2
Incomplete Nets	0
Unconnected Pads	0

Zone-Connected Nets

- +5V
- GND

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~3.26 USD
Components (estimated)	~2.23 USD
Assembly (estimated)	~2.19 USD
Total (estimated)	~7.67 USD
Batch Quantity	5
Batch Total (estimated)	~38.36 USD